Pharmacology of Indirect-Acting Cholinomimetics: Cholinesterase Inhibitors

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College of Osteopathic Medicine

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I am available to groups and individuals for pharmacology help and discussions by appointment.

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Following completion of the preparation materials, students should be able to:

1. PK properties:

Compare the class pharmacokinetic properties of the indirect-acting cholinomimetics to those of the individual agents, including the tendency to cross the blood-brain barrier.

2. Structure-activity relationship → Mechanism of action and effects:

Correlate the chemical structures of cholinesterase inhibitors with their mechanisms of action, duration of action, and cholinomimetic effects (the quaternary alcohol, carbamate esters, and organophosphates cholinesterase inhibitors).

3. Cholinomimetic effects → Therapeutic uses

Relate the effects of the cholinesterase inhibitors to their therapeutic effects and diagnostic uses.

4. Cholinomimetic effects → Toxicities and cautions for use:

Predict the signs and symptoms of cholinesterase toxicity based on the mechanisms of action and effects of the drugs.

5. Poisoning antidotes:

Explain the actions and effects of the antidotes for cholinesterase toxicity.

Preparation Materials

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Required

- **ScholarRx Bricks | Practice Questions**

Suggested supplemental resources if you wish to use them:

- Video Lecture | Dr. Goldstein's Word handout | Guided Reading Questions
- Links to a textbook and pharmacology review books are provided on the next slide.

SUGGESTIONS:

- ***Use the resources that work best for you.***
- ***You do not need to study all of them. (ScholarRx Bricks & Practice Questions are required.)***
- ***Work through the GUIDED READING QUESTIONS with pen/pencil and paper.***

Try answering them in your OWN WORDS first without looking at the readings, and then again after reviewing.

- ***Practice questions (not graded): Simple Recall and Case Vignettes***

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Preparation Materials

ScholarRx Bricks Bricks: General Pharmacology > Autonomic Drugs

Autonomic Drugs: Introduction to Autonomic Drugs > Foundations and Frameworks
https://exchange.scholarrx.com/brick/autonomic-drugs-foundations-and-frameworks

Drugs That Alter Parasympathetic Tone > Parasympathomimetics > Cholinomimetics
https://exchange.scholarrx.com/brick/cholinomimetic-agents

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 7: Cholinoceptor-Activating & Cholinesterase-Inhibiting Drugs

https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3382&sectionid=281747427

NOTE: THE REVIEW BOOKS LISTED BELOW HAVE PRACTICE QUESTIONS.

LWW Health Library, Medical Education: Lippincott's Illustrated Reviews: Pharmacology; 8e, 2023; Chapter 4: Cholinergic Agonists
https://nyit.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=253325317&bookid=3222

Access Medicine Katzung's Pharmacology: Examination & Board Review, 14e, 2024; Chapter 7: Cholinoceptor-Activating & Cholinesterase-Inhibiting Drugs

https://accessmedicine-mhmedical-com.nyit.idm.oclc.org/content.aspx?bookid=3461&sectionid=285590012

The following key points are intended help the student grasp the main ideas and concepts by organizing and summarizing the “need to know” information.

Key points may also serve as a concise reference to facilitate review.

Key points: Physiology-Pharmacology Relationship

- ***Understanding acetylcholinesterase's mechanism of action is the key to understanding the actions of the cholinesterase inhibitors.***
- ***Understanding acetylcholine's effects on the nicotinic and muscarinic receptors in the CNS, the periphery, and the neuromuscular junction is the key to understanding the effects of the cholinesterase inhibitors.***
- Cholinesterase inhibitors block the actions of acetylcholinesterase, which increases acetylcholine concentrations in the synaptic clefts and neuroeffector junctions.
- The excess acetylcholine, in turn, stimulates cholinoceptors and evokes increased responses – **cholinesterase inhibitors are amplifiers of endogenous acetylcholine**.
- Cholinesterase inhibitors also inhibit butyrylcholinesterase (BChE), which is a nonspecific enzyme that hydrolyzes many different choline esters. Other names are pseudocholinesterase and plasma cholinesterase. Note: BChE is not involved in the physiologic termination of acetylcholine.

Key points: Structure-Activity Relationship

Cholinesterase inhibitors are subcategorized according to their chemical structures, which determine their durations of action.

- Simple alcohol: Edrophonium has a quaternary ammonium cation and short duration of action.
- Carbamic acid esters: Neostigmine, pyridostigmine, and physostigmine are reversible cholinesterase inhibitors with durations of action of several hours.
  - Neostigmine and pyridostimine are synthetic compounds consisting of a quaternary ammonium cation (like acetylcholine) and a carbamate group (instead of a –COOH group). Predictably, they and are poorly absorbed and distribution to the CNS is negligible.
  - Physostigmine is a naturally occurring alkaloid, a tertiary amine, that is well absorbed and distributes to the CNS due to its greater lipid solubility.
- Organophosphates (organic derivatives of phosphoric acid) are irreversible cholinesterase inhibitors with very high lipid solubility. They were developed as insecticides that would be absorbed and distributed to the insect's CNS very rapidly. Several nerve toxins, such as sarin gas, are extremely toxic to humans.

Key points: Mechanisms of action: Inhibiting AChE increases ACh levels which enhances ACh actions at nicotinic and muscarinic receptors.

To understand the mechanism of action of the cholinesterase inhibitors, it is necessary to understand the mechanism of acetylcholine hydrolysis:

- Acetylcholine (ACh) binds to acetylcholinesterase (AChE) at two sites – the ammonium cation at the anionic site and the carboxyl group at catalytic triad of the esteratic site.
- Hydrolysis of acetylcholine releases acetate, choline, and the active enzyme.

The interactions of the three subgroups of cholinesterase inhibitors differ according to their structures:

- The simple alcohol, edrophonium, reversibly binds to the anionic site, preventing access of ACh. It's duration of action is minutes.
- The carbamate esters undergo the same process as ACh. However, the carbamoylated enzyme is more resistant to hydrolysis. The duration of action is several hours.
- The organophosphates irreversibly phosphorylate the enzyme forming an extremely stable bond. The phosphorylated enzyme undergoes aging that further strengthens the bond.

Key points: Therapeutic Uses

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System / Drugs	Therapeutic uses	Mechanism
Skeletal muscle Neostigmine, Pyridostigmine	Nondepolarizing neuromuscular blocker reversal agent Myasthenia gravis treatment	Enhance ACh levels in the neuromuscular junction Direct cholinomimetic effect on skeletal muscle (and indirect and direct effects on autonomic ganglia)
Urinary tract Neostigmine	Acute colonic pseudo-obstruction (Ogilvie syndrome)	Promotes bladder emptying by contraction of the trigone (smooth muscle)
Eye Echothiophate	Accommodative esotropia (strabismus) Reduce intraocular pressure	Ciliary muscle stimulation, potentiate accommodation, facilitate aq. humor outflow
CNS Donepezil Rivastigmine Galantamine	Alzheimer disease symptomatic treatment	Appear to increase central ACh in the basal forebrain and cortex that contribute to cognitive deficits
CNS Physostigmine	Reversal of CNS anticholinergic syndrome	Increased CNS ACh synaptic levels outcompetes anticholinergic agents at cholinergic receptors

Key Points: Adverse effects predictably result from cholinomimetic actions (stimulation) on muscarinic and nicotinic receptors.

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Cardiovascular, gastrointestinal, ***eye***, and ***skeletal muscle cholinergic effects*** are most prominent effects of cholinesterase inhibitors.

Effect	System
muscle cramps / twitching muscle weakness	Skeletal muscle
miosis, visual disturbance	Eye, facilitated accommodation
bradycardia, AV nodal block, arrythmia	Cardiac
hypotension with reflex tachycardia syncope flushing	Vascular
bronchospasm	Lung
salivation, lacrimation airway secretions GI secretions profuse sweating	Glandular secretions
abdominal cramps, borborygmi, nausea, vomiting, diarrhea	GI smooth muscle and secretions
urinary urgency / incontinence	Stimulation of micturition

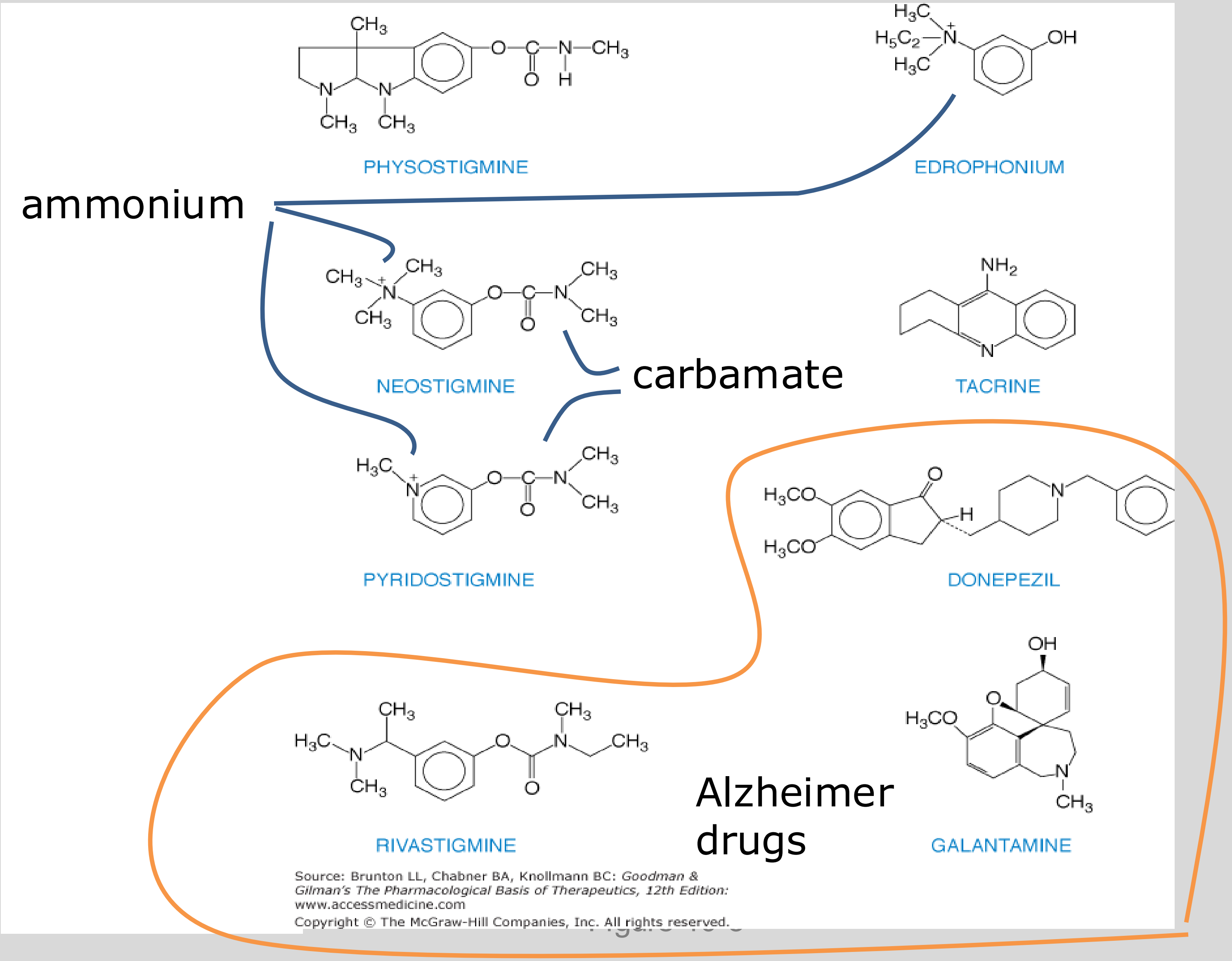
Cautions

- muscle weakness with excessive dose
- miosis may reduce night vision
- cardiovascular disease
  - – bradycardia (risk arrhythmia)
  - – heart block (slow conduction, AV node)
  - – hypotension ($\rightarrow$ reflex tachycardia)
  - – ischemic heart disease (risk angina)
  - – hyperthyroidism (risk atrial fibrillation)
- asthma
- chronic obstructive pulmonary disease
- peptic ulcer disease
- gastrointestinal or urinary bladder surgery, resection or obstruction
- Drugs that enhance adverse effects or interfere with the actions of anticholinergic drugs

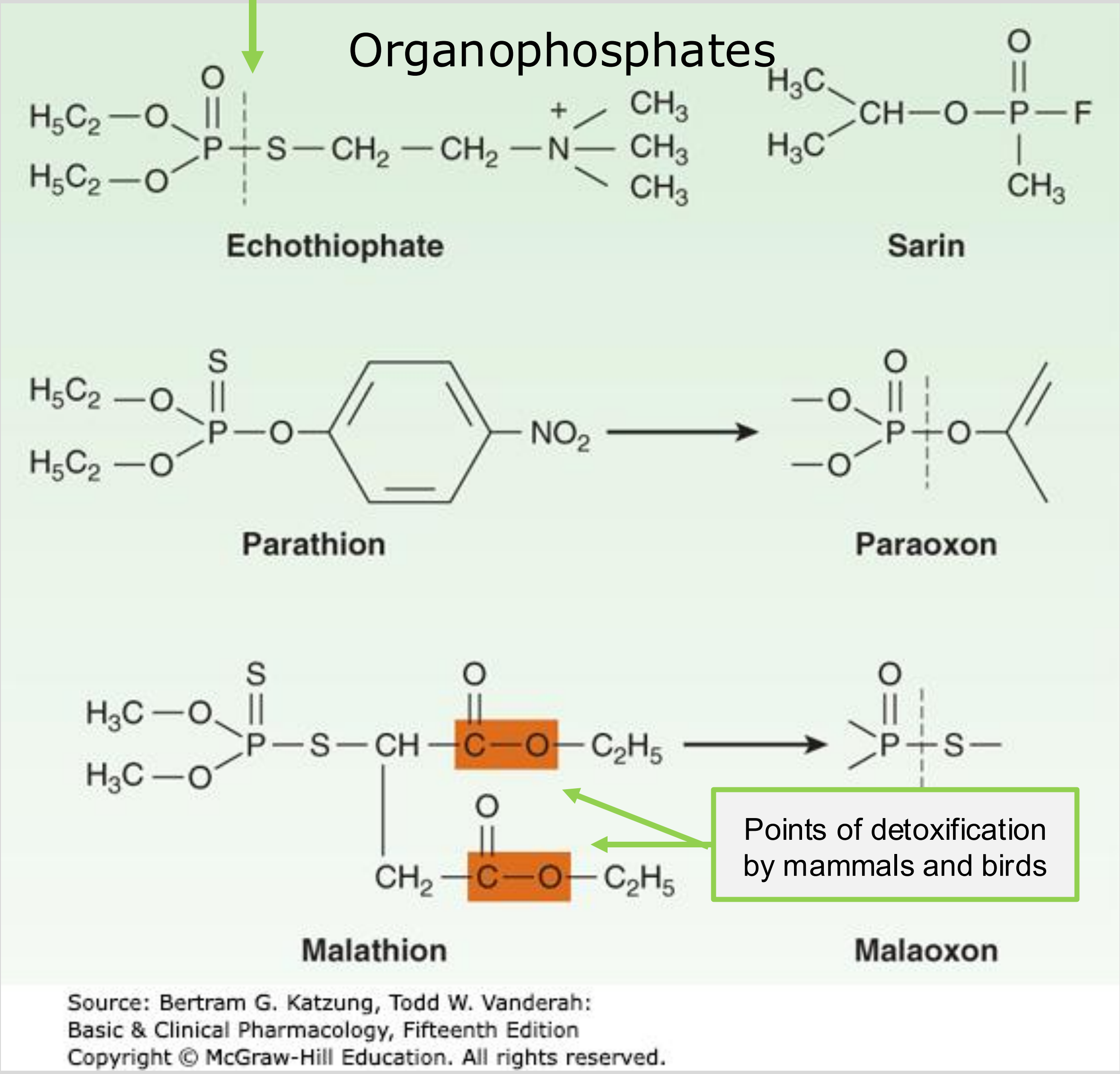
Cholinesterase inhibitors

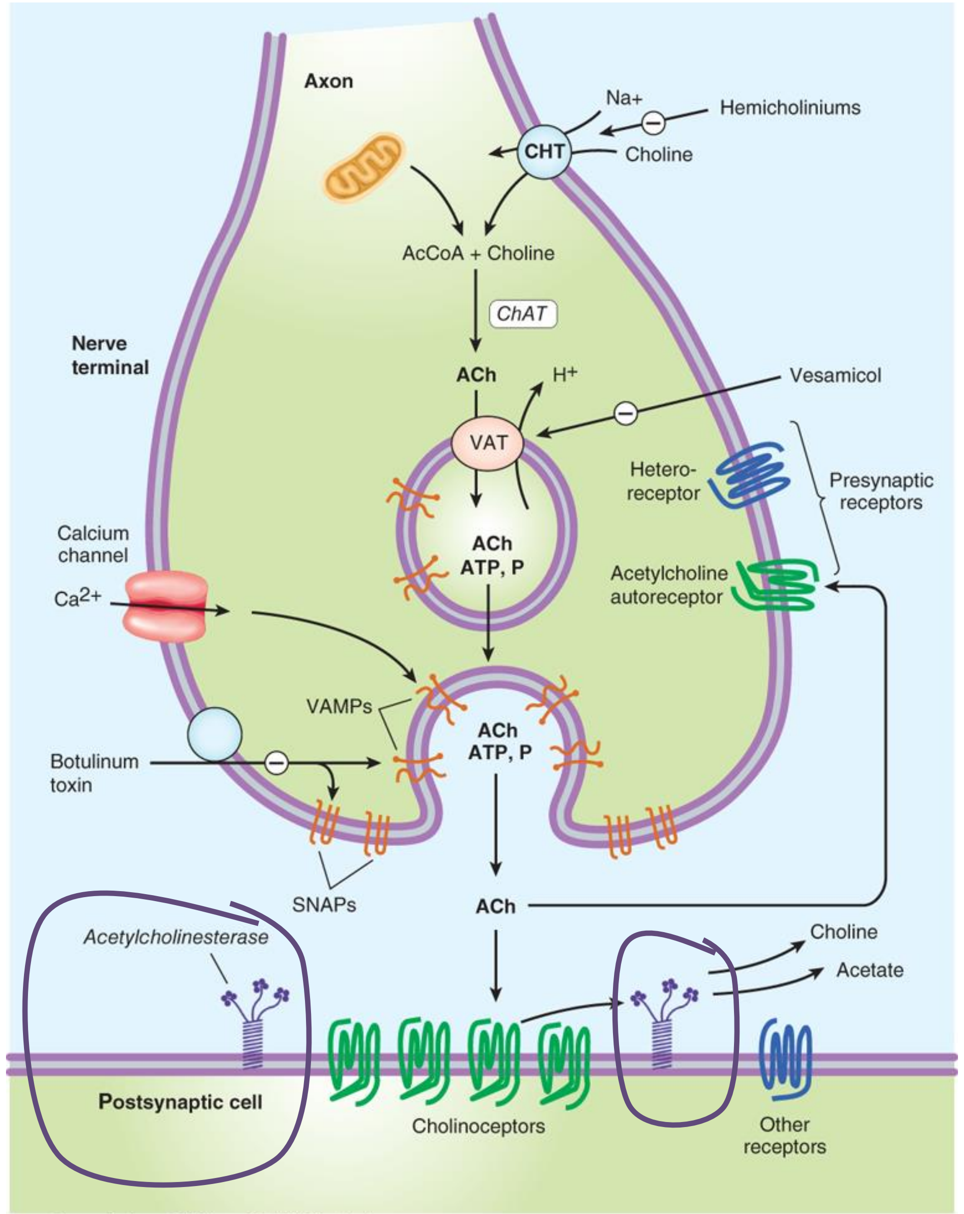
- **Indirect-acting cholinomimetics**

Agents that amplify the actions of acetylcholine



Dotted line: Bond that is hydrolyzed





1. Choline transport
2. Ch + AcCoA = ACh
3. VAT → Transport of ACh into storage vesicle
4. Axon terminal depolarization → Ca^{2+} influx
5. Vesicle attachment and fusion
6. Exocytosis of ACh & cotransmitters into junctional cleft
7. AChE → Ch + acetate → terminate action
8. Choline recycled; acetate diffuses away

Schematic illustration of a generalized cholinergic junction (not to scale). Choline is transported into the presynaptic nerve terminal by a sodium-dependent choline transporter (CHT). This transporter can be inhibited by hemicholinium drugs. In the cytoplasm, acetylcholine is synthesized from choline and acetyl-CoA (AcCoA) by the enzyme choline acetyltransferase (ChAT). Acetylcholine (ACh) is then transported into the storage vesicle by a vesicle-associated transporter (VAT), which can be inhibited by vesamicol. Peptides (P), adenosine triphosphate (ATP), and proteoglycan are also stored in the vesicle. Release of transmitters occurs when voltage-sensitive calcium channels in the terminal membrane are opened, allowing an influx of calcium. The resulting increase in intracellular calcium causes fusion of vesicles with the surface membrane and exocytotic expulsion of acetylcholine and cotransmitters into the junctional cleft (see text). This step can be blocked by botulinum toxin. Acetylcholine's action is terminated by metabolism by the enzyme acetylcholinesterase. Receptors on the presynaptic nerve ending modulate transmitter release. SNAPs, synaptosomal nerve-associated proteins; VAMPs, vesicle-associated membrane proteins.

Citation: Chapter 6 Introduction to Autonomic Pharmacology, Katzung BG, Vanderah TW. *Basic & Clinical Pharmacology, 15e;* 2021. Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=2988&sectionid=250594657 Accessed: December 03, 2021

Slide adapted by LG

Acetylcholinesterase Inhibitors: Actions and Effects

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Indirect-acting cholinomimetics

inhibit the hydrolytic action of acetylcholinesterase

↓

Acetylcholine accumulates

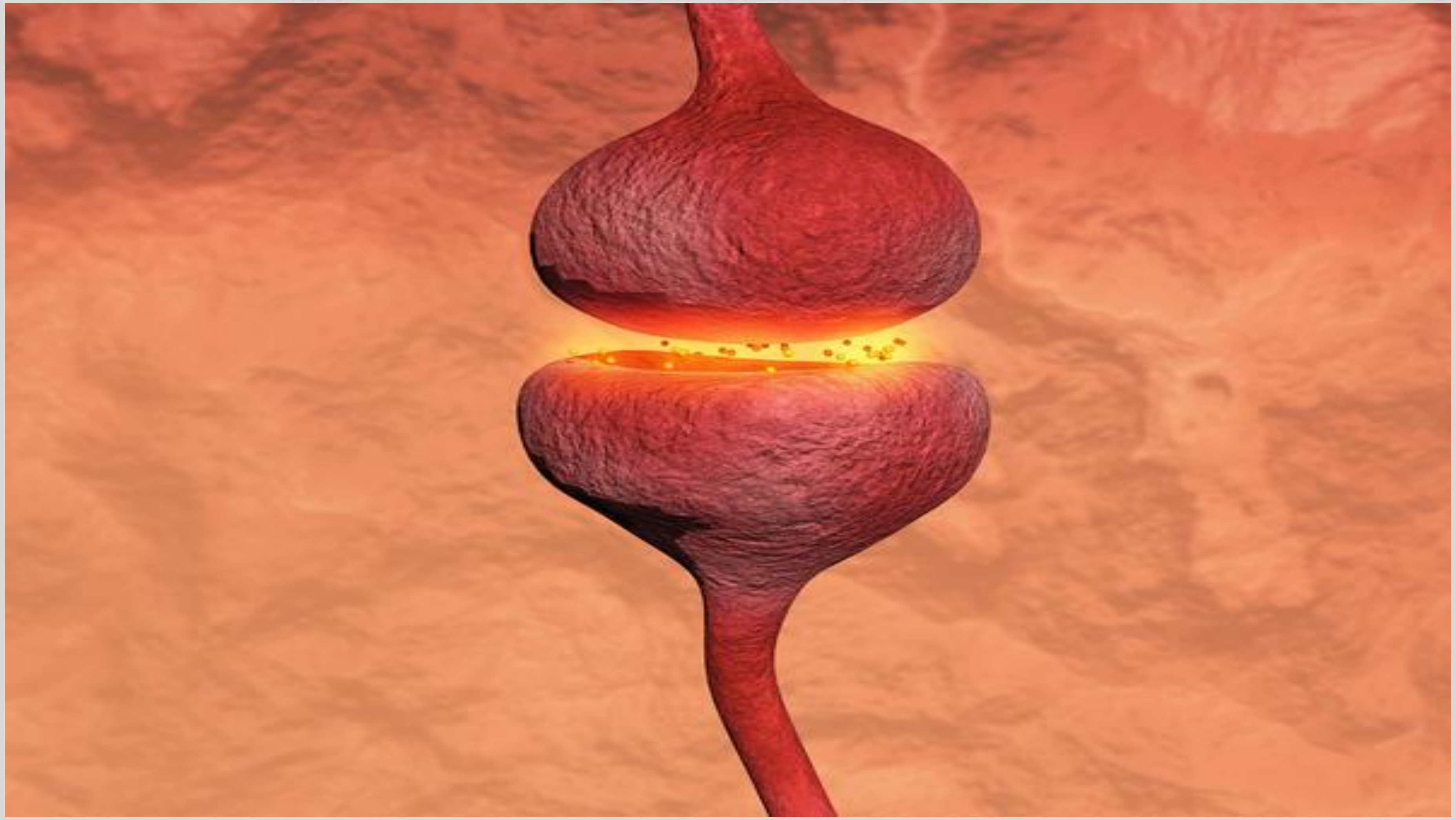
in the vicinity of cholinergic nerve terminals

↓

Enhances/Augments acetylcholine activity

Clinical pearl: Accumulation increases the potential for stimulation – even excessive stimulation – of cholinergic receptors throughout the central and peripheral nervous systems.

Preventing the breakdown of acetylcholine increases the level and duration of acetylcholine action.



- Why are cholinesterase inhibitors called indirect-acting cholinomimetics?
- What are the effects of inhibiting the actions of cholinesterase?

Figure: https://www.healthtap.com/user_questions/487889

Cholinesterase Inhibitors Subclasses

quaternary alcohol	carbamic acid esters	organophosphates
✳Edrophonium (no longer available)	✳Neostigmine	✳Malathione
	✳Pyridostigmine	✳Echothiophate
	✳Physostigmine	Insecticides
		Nerve agents

Cholinesterase Inhibitors for Symptomatic Alzheimer Disease

✳Donepezil
Rivastigmine
Galantamine

Students should know the starred (✳) drugs by name and their drug-specific features.

Pharmacokinetics of ChE Inhibitors

The light orange highlights drug-specific properties.

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	Edrophonium[!]	Neostigmine	Pyridostigmine	Physostigmine
A	IV, IM	IV, IM, SubQ	**Oral**, IV, IM	IV, IM
D	Does not penetrate CSF	Does not penetrate CSF	Does not penetrate CSF	**Readily crosses blood-brain barrier**
M	Cholinesterases	Cholinesterases	Cholinesterases	Cholinesterases
F	n/a	n/a	10-20%	n/a
E	Rapid renal excretion	Urine, ~50-80% unchanged	Urine, 80-90% unchanged	Insignificant renal excretion
		Dose adjustment for renal impairment		
t½	Duration of action 2-10 minutes (IV)	0.5-2 h Dosing is based on patient response	1-2 h **Duration 3-4 h** Dosing is based on patient response **(Oral, 5-6x /day)**	1-2 h Duration 45-60 minutes

F: oral bioavailability; Half-life: Adults with normal renal function

[!]Edrophonium is no longer available in the US. Lexidrug monographs

Mechanism: Acetylcholinesterase inactivation of acetylcholine

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Step 1

Ammonium binds to anionic site.
Carboxyl group binds serine
in the catalytic triad
of the esteratic site.

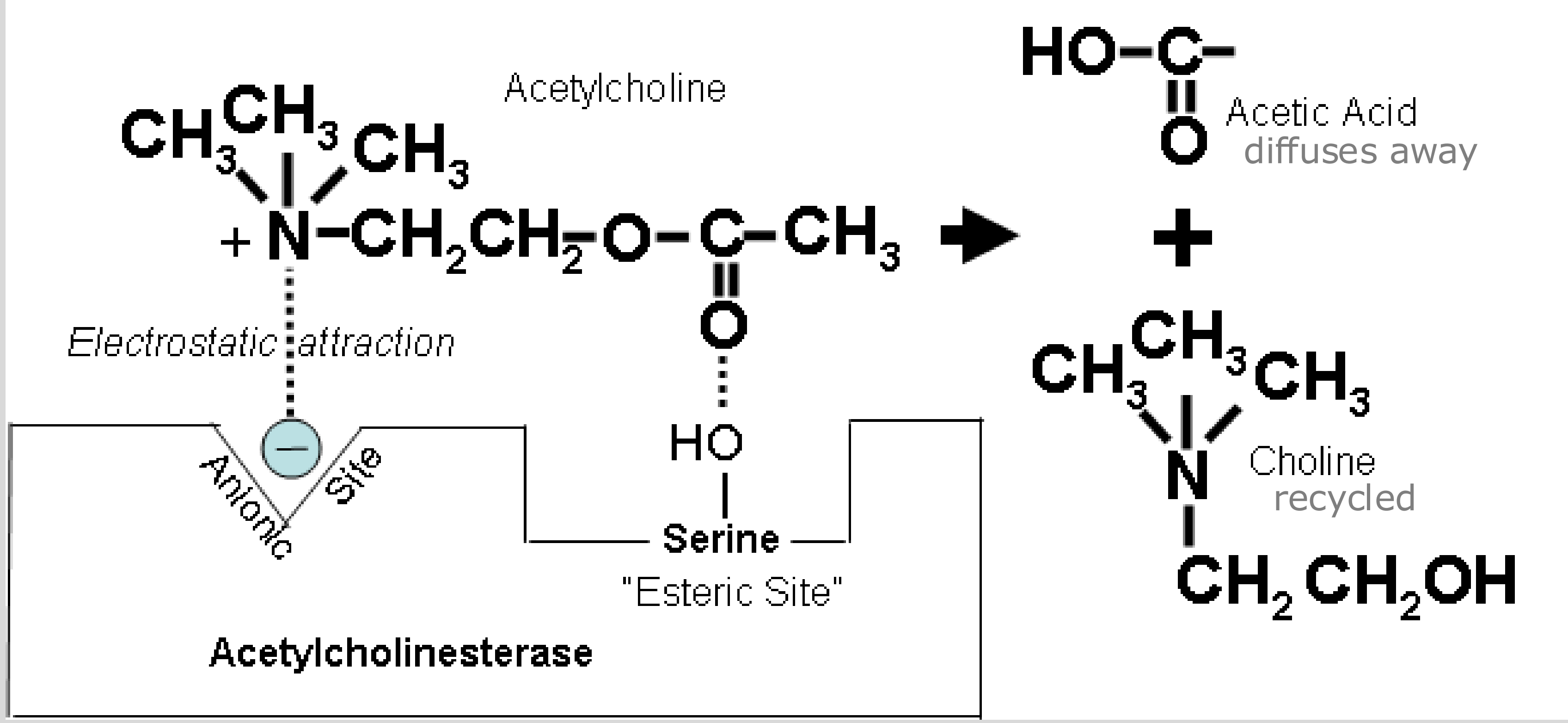
Step 2 Catalytic Step

enzyme cleaves choline
from ACh
↓ Free choline
enzyme-acetate intermediate

Step 3 Hydrolysis Step

acetate couples to the
esteratic site of the enzyme
H_2O
Bond is split ↓
acetate + active enzyme

Hydrolysis of ACh releases:
choline + acetic acid + active enzyme & termination of ACh action



Inactivation of ACh by AChE takes about 150 microseconds.

https://www.atsdr.cdc.gov/csem/cholinesterase-inhibitors/inhibitors.html

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Cholinesterase Inhibitors Mechanism of Action

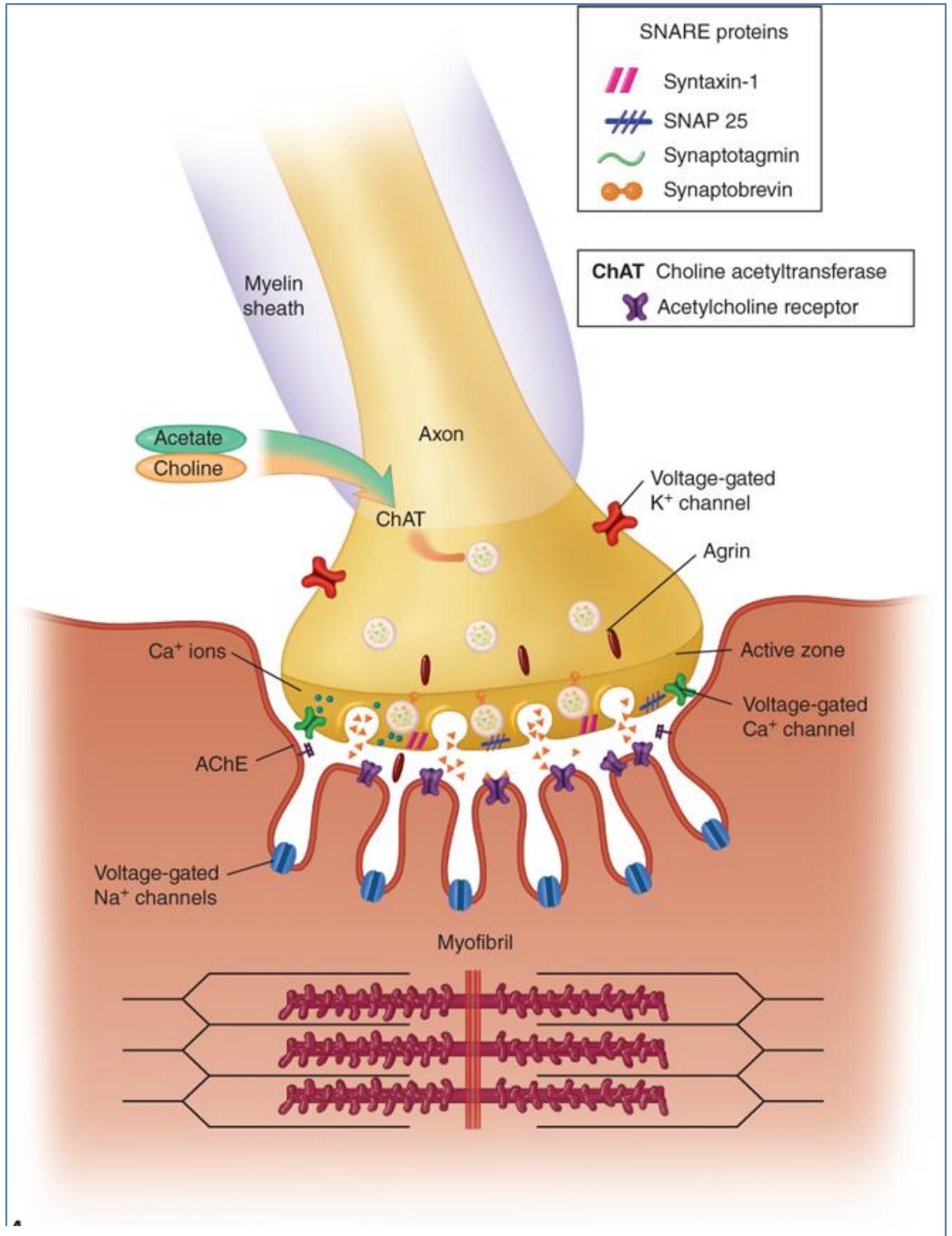
Quaternary alcohol	**Competitive Inhibition** • reversibly binds electrostatically and by hydrogen bonds to the active site • rapid dissociation → short duration of action (2-10 min)
Carbamic acid esters	**Carbamoylation** • Ammonium binds anionic site, carbamate binds esteratic site • slow hydrolysis →↑ duration of action (30 min – 6 h)
Organophosphates	**Phosphorylation** • covalent phosphorylation or the enzyme • extremely stable bond; hydrolyzes very slowly → essentially irreversible • Aging: further strengthening of phosphorus-enzyme bond → irreversible
Cholinesterase inhibitors for symptomatic treatment of Alzheimer disease (pharmacology from slide 29)	**Reversible cholinesterase inhibitors** • Donepezil: noncarbamate • Galantamine: noncarbamate • Rivastigmine: carbamate **Cross blood-brain barrier** • noncompetitive ChE inhibitor • competitive ChE inhibitor • carbamoylation of ChE

Drug	Uses	Duration of Action
Edrophonium no longer available	Dx of Myasthenia Gravis (historically) Edrophonium had suboptimal sensitivity and specificity. The most common test is serology to detect AChR autoantibodies.	~ 10 min (IV)
Neostigmine	Reversal of nondepolarizing neuromuscular blockade Myasthenia gravis treatment Acute colonic pseudo-obstruction (Ogilvie syndrome)	2.5-4 hrs
Pyridostigmine	Myasthenia gravis treatment Reversal of nondepolarizing neuromuscular block	6-8 hrs

Neostigmine and pyridostigmine (1) ***inhibit cholinesterase*** and (2) have ***direct nicotinic agonist effect*** at the N_M junction and autonomic ganglia.

Physostigmine	**Antidote for anticholinergic toxicity** Reversal of toxic, life-threatening delirium caused by anticholinergic agents (e.g. atropine, jimson weed). **Toxicities → cautious use:** Physostigmine can cause bradycardia, respiratory distress, and seizures with too rapid administration.	45-60 min
Malathion	Topical lotion: head and pubic lice and ova, scabies	Lotion must remain on skin 8-12 h
Echothiophate	Ophthalmic drops: miotic, reduction of intraocular pressure	Long duration

Myasthenia Gravis pathophysiology



a normal presynapatic N_M junctions & a normal postsynapatic terminal

Autoimmune response

↓

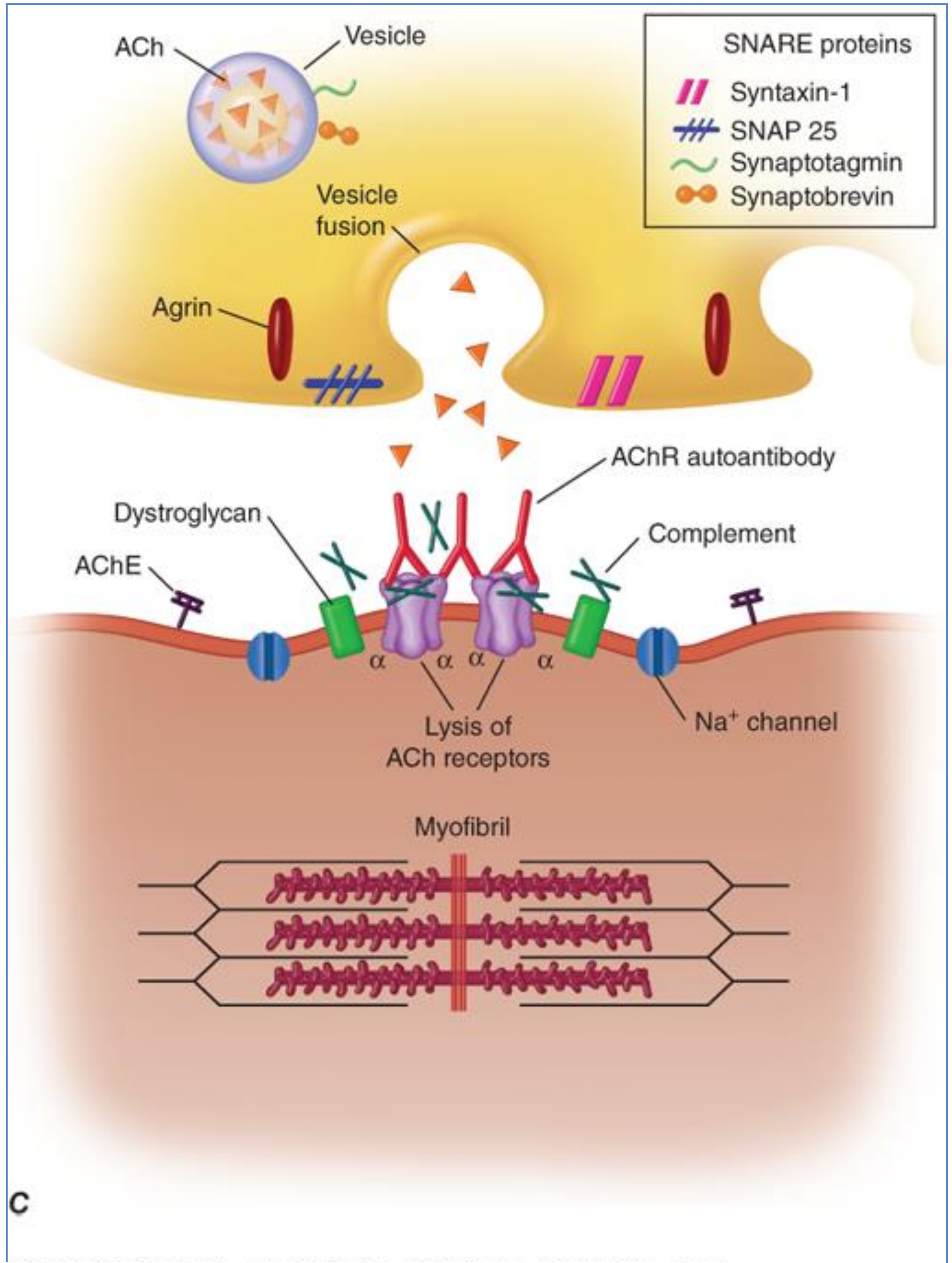
anti-nAChR antibodies

↓

cross-linking &
lysis of nAChRs
which decreases the number of receptors at the N_M junction

↓

failure to trigger muscle action potential (end-plate potential)



normal nerve terminal; ↓number of nAChRs (stippling); flattened, simplified postsynaptic folds; and a widened synaptic space

Mention:

Edrophonium test for differentiating myasthenic crisis and cholinergic crisis is no longer performed but **it may still be on the board exams**.

Definitions:

- Myasthenic crisis: Life-threatening respiratory failure due to weakness of respiratory muscles.
- Cholinergic crisis: Excessive cholinesterase inhibitor intake overstimulates nicotinic receptors causing depolarizing block resulting in muscle weakness and respiratory failure.

Cholinergic crisis occurs very rarely if at all in patients taking pyridostigmine at therapeutic doses.

- **Practice point:** Detailed history and physical examination are necessary to diagnose cholinergic crisis. A history of exposure to cholinesterase inhibitors is key to diagnosing cholinesterase inhibitor toxicity.

The Test: A small dose of edrophonium $\rightarrow$ brief block of AChE in the N_M junction

(1) Improvement of muscle weakness is a positive indicator of myasthenia gravis.

(2) Muscle weakness does not resolve or worsens in cholinegeric crisis.

Check your knowledge

- A patient who had surgery under general anesthesia facilitated by a nondepolarizing neuromuscular blocker is administered a drug to reverse the neuromuscular blockade. What is the mechanistic rationale for this approach?

- A patient diagnosed with myasthenia gravis is given treatment with an oral drug. What is the drug and what is the clinical rationale for this approach?

Acute Cholinergic Toxicity

- Caused by exposure to organophosphates and carbamates

Carbamate inhibitors are used to control insects, weeds, and fungi on food crops, lawns and forests. Organophosphate insecticides are still available.

Cholinergic Toxidrome: cholinergic excess signs / symptoms

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DUMBELS

- **D**iarrhea
- **U**rination
- **M**iosis
- **B**ronchoconstriction
- **B**ronchorrhea
- **B**radycardia
- **E**mesis
- **L**acrimation
- **S**alivation
- **S**weating (diaphoresis)

drip, drip, drip

MTWtHF

(days of the week)

- **M**ydriasis
- **T**achycardia
- **W**eakness
- **H**ypertension
- **F**asciculations

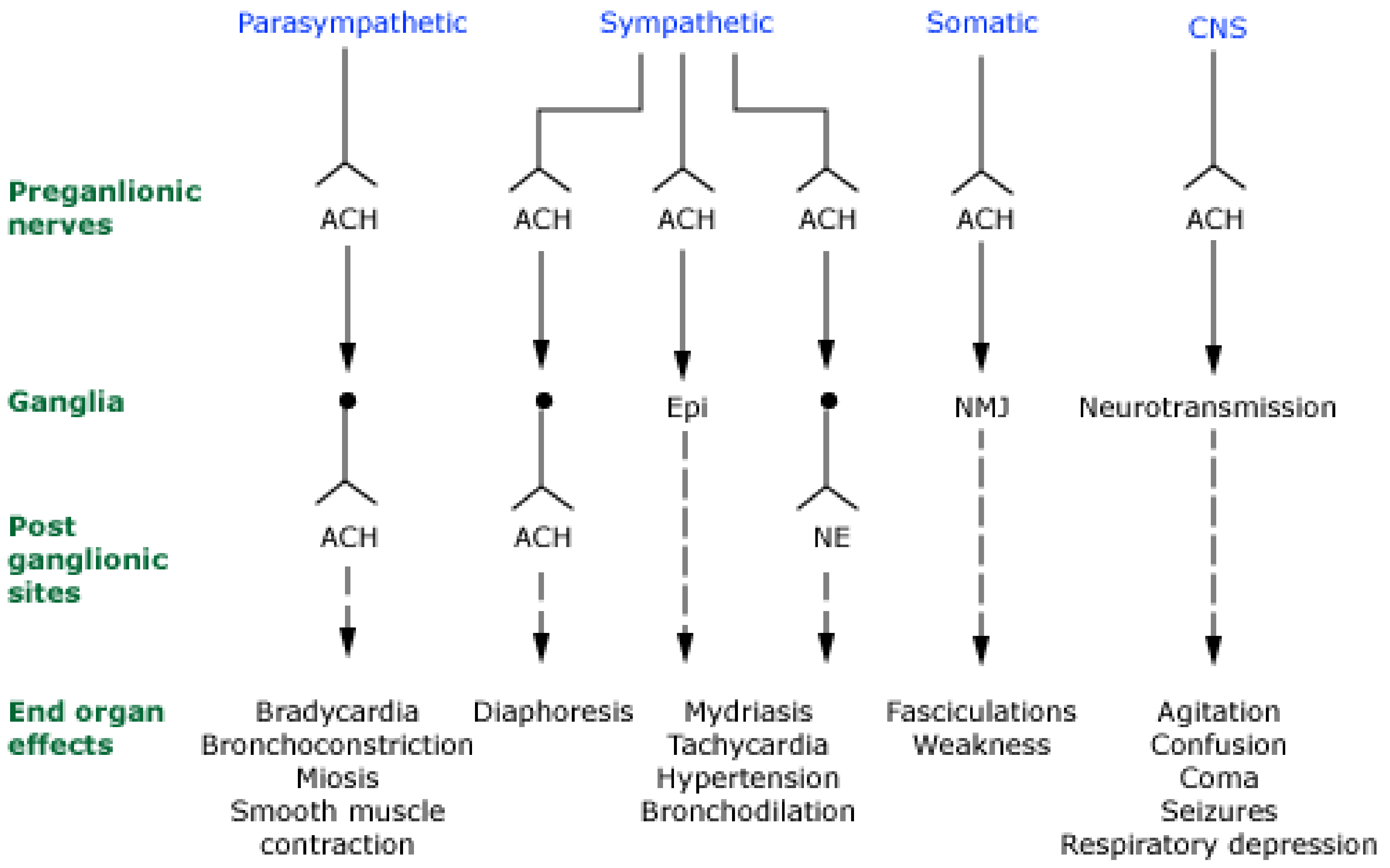
CNS

- Central respiratory depression
- Lethargy
- Seizures
- Coma

Death results from respiratory failure:

Central respiratory depression
Bronchoconstriction
Bronchorrhea
Weakness and paralysis of respiratory muscles.

Neurologic effects of organophosphate and carbamate agents



ACH: acetylcholine; Epi: epinephrine; NE: norepinephrine; NMJ: neuromuscular junction.

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CH: acetylcholine; Epi: epinephrine; NE: norepinephrine; NMJ: neuromuscular junction

Acute Organophosphate Toxicity

Diisopropylfluorophosphate (industrial uses) / Insecticides (*eg* diazinon, malathione, others)

Nerve agents (soman, sarin, VX, others)

- **Manifestations:** muscarinic, nicotinic, and CNS effects
- **Onset:** minutes (inhaled); delayed after GI or percutaneous absorption
- **Duration:** determined by the properties of the compound
- **Diagnosis:** H/O exposure; S/S; measurement of cholinesterase activity in erythrocytes and plasma

Management:

- Removal of toxic agent
- Respiratory and cardiovascular support
- Treat seizures

Antidotes:

- **Atropine:** antimuscarinic agent
- **Pralidoxime (2-PAM)** Cholinesterase regenerator (strong nucleophile): may relieve respiratory muscle paralysis

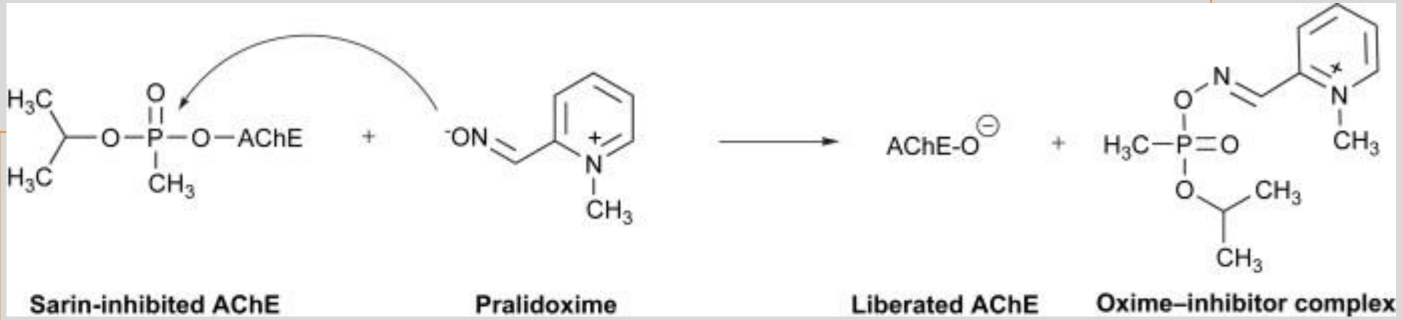
1) Atropine: Antimuscarinic Agent
2) Pralidoxime (2-PAM): Cholinesterase Regenerator

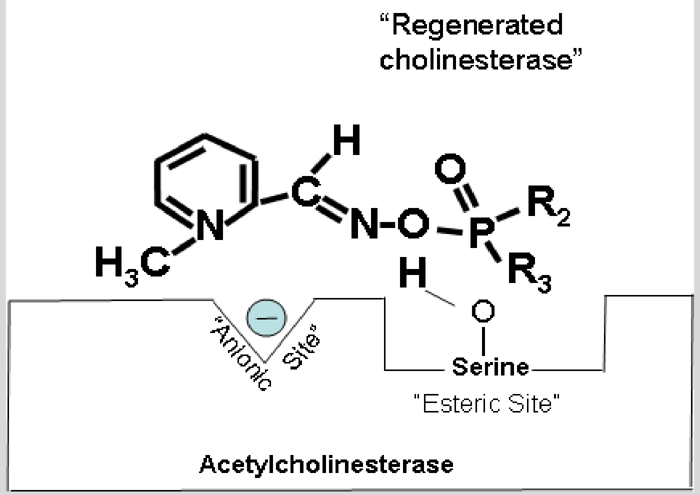
Atropine

- Competitive antagonist at muscarinic receptors
- ***Reduces the symptoms of parasympathetic overstimulation*** resulting from excess acetylcholine in the synapses
- **Peripheral and CNS effects**

Pralidoxime

- Strong nucleophile
- High affinity for phosphorus atom in organophosphate
- Breaks the phosphorus-enzyme bond
- **Peripheral effects only**





Pralidoxime breaks the bond IF administered before aging occurs.

Interaction: https://www.sciencedirect.com/topics/neuroscience/pralidoxime

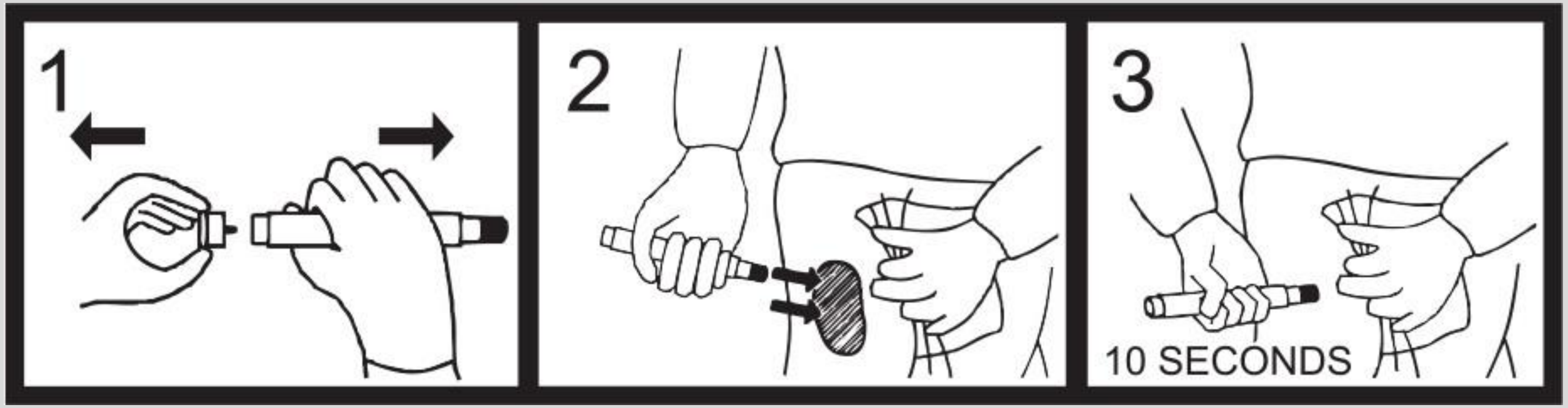
Diagram: Agency for Toxic Substances and Disease Registry Environmental Health and Medicine Education; Cholinesterase inhibitors http://www.atsdr.cdc.gov/csem/csem.asp?csem=11&po=0

Nerve Agent, Auto-Injector (ATNAA)

Atropine Injection and Pralidoxime Chloride Injection

in separate chambers as sterile, pyrogen-free solutions for intramuscular injection.

self-aid or buddy-aid



https://fda.report/DailyMed/596c7a8f-27cd-4de2-9491-476f43570b8b#google_vignette

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Check your knowledge

- The carbamate, organophosphate, Alzheimer’s cholinesterase inhibitors have myriad, predictable, mechanism-related adverse effects. What systems are involved?

Centrally-Acting Cholinesterase Inhibitors

for the symptomatic treatment of Alzheimer disease

- CNS actions and effects for the symptomatic treatment of Alzheimer disease:
  - Cholinergic neurons project to various cortical areas. These neurons control attention, thinking, and processing of stimuli. They span the forebrain and encompass the brainstem and thalamus, which are responsible for consciousness and attention.
  - In Alzheimer dementia, these cholinergic neurons have defective projections that correlate with the classic symptoms of cognitive slowing and decline.
  - Cholinesterase inhibitors enhance synaptic levels of ACh.

Centrally-Acting Cholinesterase Inhibitors

Drug	PK (all oral)	MOA (probable)
Donepezil	• **oral; well absorbed; 1x daily dosing** • transdermal option (1x/wk) • protein binding >95% • hepatic (minor CYP), active metabolites • **100% bioavailability** • Urine mainly / feces: **renal dosing not required** • **t½ ~70h**	reversible • noncompetitive inhibition
Rivastigmine	• 35% bioavailability, cholinesterase hydrolysis, renal (metabolites), duration of activity ~10h	reversible • carbamoylation
Galantamine	• 90% bioavailability; hepatic metabolism (CYP2D6, 3A4); renal, biliary excretion; t½ 7h	reversible • competitive inhibitor
Tacrine	• **no longer used due to hepatoxocity** • oral; t½ 2-4h; admin 4x daily before meals / inhibits AChE and BuChE; and other effects	

These drugs are generally well tolerated.
Most common side effects: nausea, anorexia, vomiting, diarrhea, altered sleep patterns, vivid dreams
Potentially serious (may require deprescribing):
bradycardia, hypotension, syncope, AV block, unintentional weight loss

Check your knowledge

- What are the PK properties of donepezil and their importance in clinical management of the patient with Alzheimer disease?

- What are the common side effects?

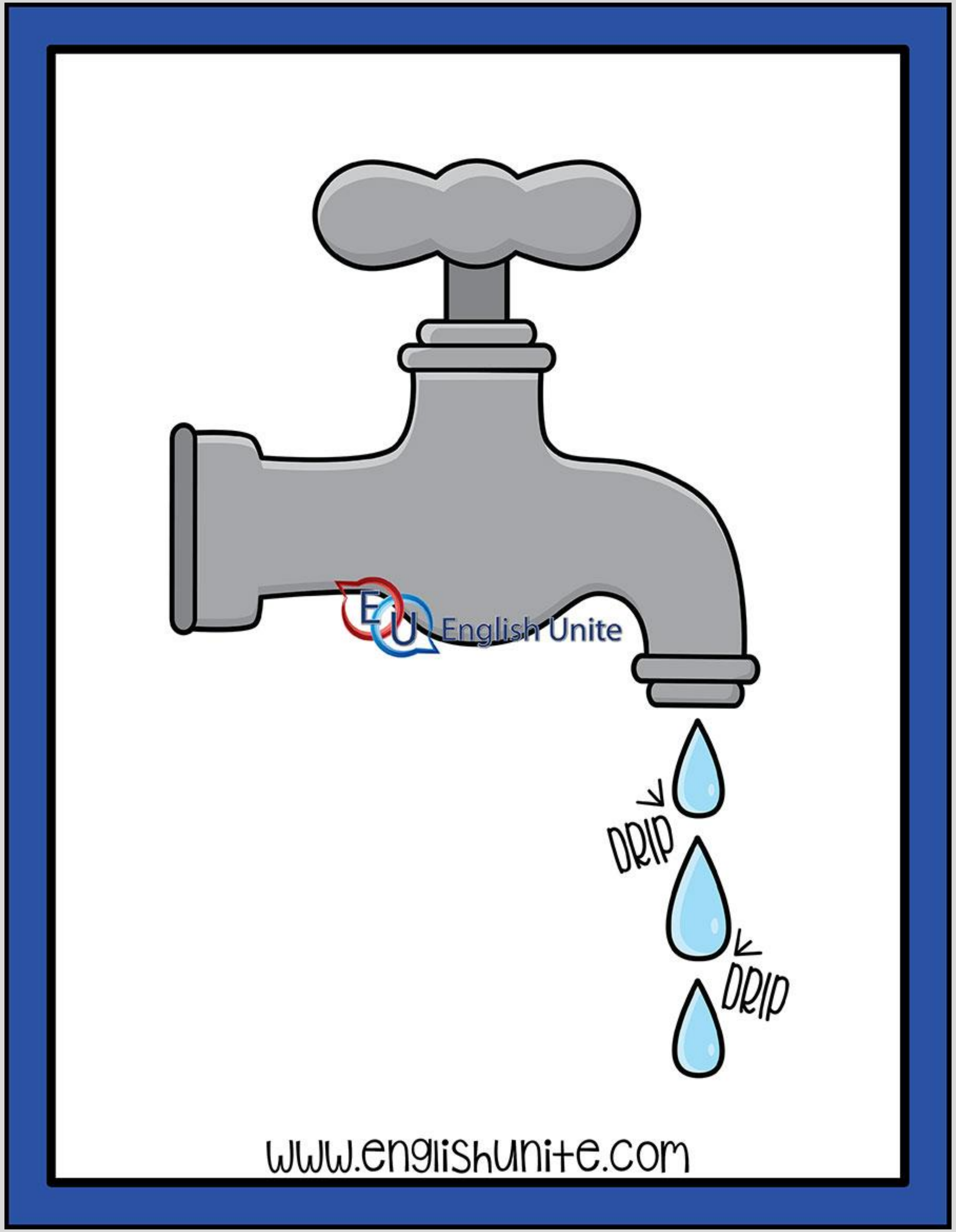
- What adverse effects might require deprescribing of the drug?

Summary

- Cholinesterase inhibitors amplify the actions and effects of acetylcholine by inhibiting ACh breakdown, which increases ACh concentration and residence time in cholinergic synapses.
- Cholinesterase inhibitors have both nicotinic and muscarinic effects. Dominant effects vary among organ systems. Cardiovascular, gastrointestinal, eye, and skeletal muscle cholinergic effects are most prominent effects of cholinesterase inhibitors.
- Edrophonium is a simple alcohol that reversibly binds the active site electrostatically and by hydrogen bonds. It has with a very short duration of action.
- Carbamate esters undergo are much more resistant to hydrolysis than ACh. Thus, they have a longer duration of action.
- Organophosphates form an extremely stable phosphate-enzyme bond. Aging further stabilizes bond (irreversible). Organophosphate toxicity can be rapidly fatal.
- Adverse effects of cholinesterase inhibitors reflect the actions of acetylcholine on the various organ systems: diaphoresis, salivation, abdominal pain, nausea, vomiting, diarrhea, urination, bronchoconstriction and secretion of mucus, bradycardia, hypotension, shortness of breath (bronchoconstriction, bronchorrhea), respiratory depression, and respiratory paralysis.
- Atropine is a muscarinic receptor antagonist, which blocks acetylcholine's effects in periphery and CNS.
- Pralidoxime (2-PAM) is a strong nucleophile with a very high affinity for the phosphorus atom in organophosphates that regenerates acetylcholinesterase before aging occurs, but not afterwards. It does not penetrate into the CNS. The two drugs together may reverse both muscarinic and nicotinic signs and symptoms.

Summary: Therapeutic Uses of Cholinomimetics (direct- and indirect-acting)

Alzheimer disease Symptomatic treatment	Donepezil; Rivastigmine; Galantamine
Anticholinergic toxicity Reversal of toxic life-threatening effects	Physostigmine
Myasthenia gravis symptom relief	Pyridostigmine (oral) Neostigmine (injection)
Reversal of neuromuscular blockade	Neostigmine; Pyridostigmine
Urinary retention due to neurogenic atony of the urinary bladder; post-operative, post-partum	Bethanechol
Acute colonic pseudo-obstruction (Ogilvie syndrome)	Neostigmine
Pediculocide/ovicide, head and pubic lice; scabies	Malathione topical lotion
Reduction of elevated intraocular pressure	Pilocarpine; Carbachol Echothiophate
Smoking cessation	Nicotine; Varenicline
Xerostomia: Dry mouth 2° head and neck surgery or Sjogren's syndrome (if salivary parenchyma retains residual function)	Pilocarpine; Cevimeline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Check your knowledge

- A patient who had surgery under general anesthesia facilitated by a nondepolarizing neuromuscular blocker is administered a drug to reverse the neuromuscular blockade. What is the mechanistic rationale for this approach?
- Administration of a carbamate cholinesterase inhibitor (neostigmine or pyridostigmine) reversibly inhibits acetylcholinesterase, which increases ACh synaptic levels. ACh outcompetes the neuromuscular blocker, reversing the blockade.
- A patient diagnosed with myasthenia gravis is given treatment with an oral drug. What is the drug and what is the clinical rationale for this approach?
- Myasthenia gravis is characterized by autoimmune attack on neuromuscular nicotinic receptors, cross-linking them and reducing their numbers. Pyridostigmine inhibits AChE, increasing synaptic ACh levels, which increases the probability of producing an action potential, leading to variable improvement in muscle strength.

Check your knowledge

- The carbamate, organophosphate, Alzheimer's cholinesterase inhibitors have myriad, predictable, mechanism-related adverse effects. What systems are involved?
- All autonomic nervous system functions can be activated because ACh is the neurotransmitter at the nicotinic receptors of the ganglia (PNS and SNS), the nicotinic receptor at the adrenal medulla (SNS), the nicotinic receptors at the neuromuscular junction, and the muscarinic receptors at the effector organs (PNS except sweat glands, SNS).
- Neostigmine and pyridostigmine do not cross the BBB.
- Physostigmine, donepezil, and other Alzheimer disease drugs cross the BBB so have effects on the CNS and peripheral nicotinic and muscarinic receptors.

Check your knowledge

- What are the PK properties of donepezil and their importance in clinical management of the patient with Alzheimer disease?
- Oral – easy for the patient to take and the care giver to administer.
- Once weekly transdermal option, if preferable, but VERY expensive
- 100% bioavailability – reliable blood levels
- Excreted mainly in urine mainly as metabolites – dosing adjustment for renal insufficiency is not required
- Long half-life (~70 h) – once daily dosing
- What are the common side effects?
- nausea, anorexia, vomiting, diarrhea, altered sleep patterns, vivid dreams
- What adverse effects might require deprescribing of the drug?
- bradycardia, hypotension, syncope, AV block, unintentional weight loss

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